WiFi Feature Engineering Pipeline Explanation

1. Objective
Convert raw WiFi device logs into a structured, time-series dataset suitable for anomaly detection using machine learning.

2. Nature of Raw Data
The input data consists of cumulative counters collected at fixed intervals (e.g., every 10 seconds). These counters continuously increase and cannot be directly used for analysis.

3. Time-Series Formation
Data is sorted by device (client_mac) and timestamp. The difference between consecutive timestamps is computed to establish temporal continuity.

4. Counter to Rate Conversion
Cumulative counters are transformed into rates using differencing. For example, packet rate = (current - previous) / time interval. This converts static counters into dynamic behavioral signals.

5. Activity Detection
An 'is_active' flag is created to distinguish meaningful traffic intervals from idle periods where no packets are transmitted or received.

6. Reliability Metrics
Features such as tx_success_rate, retry_rate, tx_failed_rate, and rx_error_rate quantify transmission reliability and network health.

7. Throughput Features
Transmission and reception rates are converted into Mbps to represent data transfer efficiency.

8. PHY Layer Features (MCS)
MCS (Modulation Coding Scheme) features capture link quality. Ratios (low/mid/high), weighted score, and entropy describe signal strength and stability.

9. Security Metrics
Security-related errors (MIC, PN, decryption) are aggregated into a single feature indicating protocol-level anomalies.

10. Data Cleaning
Invalid values (NaN, infinity) are handled to ensure compatibility with machine learning models.

11. Final Dataset
Each row represents the state of a device at a given time interval. The dataset is now structured as a time-series feature matrix ready for anomaly detection.

12. Key Insight
The pipeline transforms raw logs into meaningful behavioral signals, enabling unsupervised models to learn normal patterns and detect deviations.